

SPHinX Authentication Technique: Secure Painting authentication in eXtended reality

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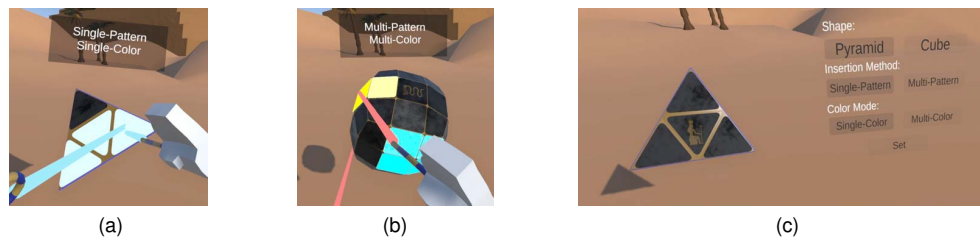


Figure 1: The SPHinX technique: interaction with a) a pyramid-shaped object for tracing a single unlocking pattern, b) a cube-shaped object for tracing multiple-colored unlocking patterns, and c) the interface for selecting the authentication method.

ABSTRACT

Authentication in immersive virtual reality, and more in general, into extended reality applications is a growing need. The crude adoption of 2D authentication schemes into virtual environments could raise security and usability issues. This paper proposes SPHinX, a novel three-dimensional (3D) authentication scheme based on unlocking patterns. Users can authenticate by painting (tracing) patterns on a 3D object. In this way, 3D interactions can be leveraged to make the authentication scheme less vulnerable to different types of attacks, e.g., shoulder surfing. A user study was carried out by involving 16 participants with the aim to compare a number of alternatives concerning the shape of the object to be painted, the use of single or multiple patterns, and multi-factor authentication.

1 INTRODUCTION AND RELATED WORK

The key role that is envisaged for the Metaverse in our everyday life has led to the belief that authentication into immersive, co-presence environments represent key issues for the development of this technology [3]. As a matter of example, in the context of immersive applications that allow users to make in-app purchases, it is essential to verify their identity and protect personal information [4]. Several techniques have been proposed in the literature, e.g., based on the use of biometric data (voice, iris, etc.). However, data protection represents a critical aspect of these approaches, as compromising this information can lead to privacy issues [3]. Alternative strategies consist in using, in immersive environments, existing methods originally developed for 2D environments. For instance, there are works that propose the use of 2D sliding patterns or 2D Personal Identification Numbers (PINs) for authentication.

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Even though they are extremely easy to use, 2D authentication methods are also prone to attacks like, e.g., shoulder surfing [4]. For this reason, researchers are turning their attention to approaches based on 3D input, which are typically more secure than 2D ones but characterized by a lower usability [6]. As a matter of example, the work in [4] proposes a method that allows users to authenticate by inserting a PIN on the faces of a Rubik's cube. This approach proved to be a good compromise between usability and safety (e.g., in terms of resistance to shoulder surfing attacks).

Moving from the above considerations, this paper looks at the possible benefits of transposing to virtual reality (VR) the authentication paradigm based on *unlocking patterns*, since they were proven to be easier to memorize and use than PINs [1]. Although previous studies raised concerns about the vulnerability of unlocking patterns to shoulder surfing attacks when using this approach on mobile phones [2], it was demonstrated that security issues are mitigated when this technique is brought to VR [5]. More specifically, comparing the mobile version with the VR version, it was found that the latter is more resistant to the mentioned attacks and shows comparable performance in terms of usability [5]. Differently than the work in [5], the authentication method proposed in this paper leverages the three-dimensionality of virtual environments, by letting the users specify the pattern on 3D shapes rather than using a 2D interface. The impact of the shape, of patterns possibly made up of sub-patterns, and of colors for multi-factor authentication is investigated through a user study.

2 THE SPHINX TECHNIQUE

The proposed technique allows a user to authenticate in an immersive application by tracing patterns on a 3D object. So far, a pyramid (Figure 1a) and a cube (Figure 1b) have been considered.

At the beginning of the process, the shape appears as a floating object in front of the user with a random orientation (this choice was made to improve security against shoulder surfing attacks). The user can use the controller buttons to freely move the object around in the 3D space or rotate it around a fixed point. Unique symbols depicted on each face help the user to identify the proper orientation for tracing the patterns. Each face is split into a number of grid positions (six for the pyramid, nine for the cube) that are

connected to form a pattern. Patterns are traced by pressing the trigger button of one of the controllers and pointing at one of the grid positions on an object face. The user can cancel the last selection by simply returning to the previous grid position while tracing the pattern. Once the pattern is traced, the selected grid positions remain highlighted showing the input used for the authentication. Two tracing methods are supported, leveraging *single* (Figure 1a) and *multiple* (Figure 1b) patterns, respectively. In the first method, the pattern has to be traced all at once; when the user releases the trigger button, the traced pattern is evaluated regardless of its length (number of highlighted grid positions). The second method allows the user to split the pattern into a series of sub-patterns that have to be traced in a specific order; once the tracing is completed, the sub-patterns are concatenated to obtain the final pattern to be evaluated. Both the tracing methods support overlapping patterns, i.e., patterns in which a grid position is selected multiple times. In order to support multi-factor authentication, the user has the possibility to use different colors (Figure 1b), thus generating more complex patterns/sub-patterns. The color can be selected from a color palette (three colors have been used for experimental purposes).

The SPHinX technique was integrated in a Unity application (shown in Fig. 1c) compatible with the OpenXR standard. Given the acronym, a cultural heritage application was chosen, which allows the authenticated users to visit an Egyptian-themed exhibit. A video showing a user performing the authentication is available at <https://bit.ly/3Wp90Pr>.

3 USABILITY EVALUATION

In order to study the usability of the proposed technique, an experimental evaluation was carried out by involving 16 volunteers (nine male and six female) aged between 21 and 31 ($\bar{x} = 24.94$, $sd = 2.90$). After filling in a demographic questionnaire, the participants were left free to experience the SPHinX technique in a dedicated training session. Afterwards, they were requested to authenticate by using the following configurations: single pattern, no colors (SP), multi-pattern, no colors (MPnC), and multi-pattern with colors (MP). Each configuration was repeated four times (at each iteration, the orientation of the object was reset), and experimented with both the pyramid and the cube. Thus, the total amount of trials was 24 (3 configurations $\times$ 4 repetitions $\times$ 2 shapes).

The configurations to start with and the order of execution were randomly selected in order to limit possible learning effects. In all the configurations, the pattern was set to have a length equal to eight grid positions. By grounding on the findings of [4], indicating that a higher security against shoulder-surfing attacks is obtained when inserting PINs spanning multiple faces of a 3D object, it was chosen to work with four-sided patterns for the evaluation.

During the experiments, two measures were collected with the aim of comparing the considered configurations: i) *login speed*, i.e., the time required by the participants to trace a pattern, from the first to the last grid position [5], and ii) *error rate*, the number of incorrect patterns traced by the participants [5]. Furthermore, the participants were asked to fill in a post-test questionnaire based on the statements proposed in [5]. In particular, the questionnaire was aimed to evaluate the configurations in terms of ease of use, perception of the operation speed, likeability of the configuration, and ease of error recovery. Each statement was evaluated using a 1-to-5 Likert scale (from very low to very high).

4 RESULTS

The statistical significance of the differences was analyzed with the Friedman test, whereas pairwise comparisons were studied with the Wilcoxon test for paired samples. Considering statistically significant differences and starting from the login speed (reported by making reference to the average time for tracing a pattern over a grid position, as done in [4]), all the configurations using the pyramid

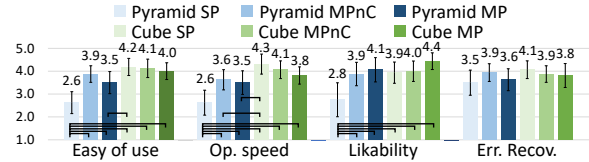


Figure 2: Usability evaluation based on statements reported in [5].

object (SP: 1.99s, MPnC: 2.14s, and MP:1.81s) were slower than those using the cube object (SP: 1.10s, MPnC: 1.05s, and MP:1.15s $p < .001$). No statistically significant differences were identified for the error rate. The objective outcomes were also confirmed by the subjective results reported in Fig. 2 (significant differences are indicated using horizontal whiskers). In particular, it is possible to observe that the configurations leveraging the cube were judged as easier to use ($p < .001$), faster ($p < .001$), and more liked ($p = .004$) by the participants than those using the pyramids object.

5 CONCLUSIONS AND FUTURE WORK

This paper proposed an authentication technique based on unlocking patterns. Different configurations were compared, by leveraging both objective and subjective measurements. The study specifically focused on usability and involved 16 volunteers. The obtained results showed the superiority of the cube with respect to the pyramid as 3D interface for tracing the patterns in terms of login speed and usability factors. The lack of statistically significant differences made it not possible to identify a clear rank among the tested configurations.

Future work will be aimed to study the robustness of the proposed technique to different types of attacks. It is also planned to extend the analysis by considering other 3D shapes and evaluating the impact of using a different number of grid positions. The cognitive effort associated with pattern tracing will also be considered, with the goal of better quantifying the tradeoff between security and user experience. After a comprehensive study of the possible configurations that can be supported by the devised approach, users could be provided with the possibility to choose the authentication configuration they prefer based on different aspects, such as memorability, robustness, perceived operation speed, etc.

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